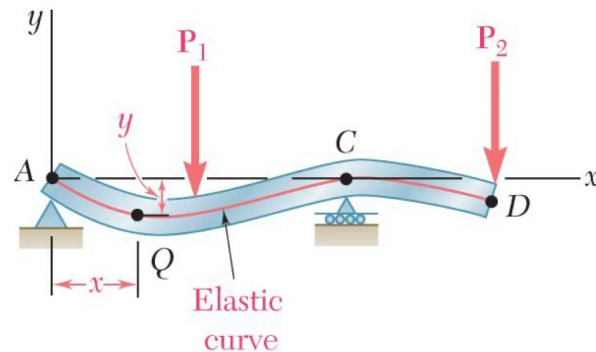


Deflection Curve Functions

Method of Integration

$$y''(x) = \frac{M(x)}{EI}$$



$$y^{(3)}(x) = \frac{V(x)}{EI}$$

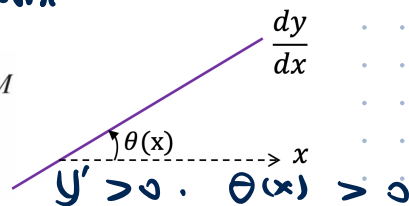
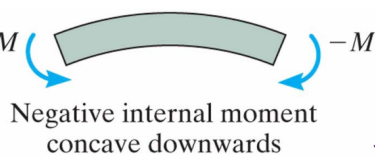
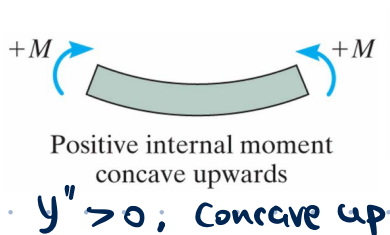
$$y^{(4)}(x) = \frac{W(x)}{EI}$$

$y(x)$: Curve function $y'(x)$: Curve slope function

Sign Conventions

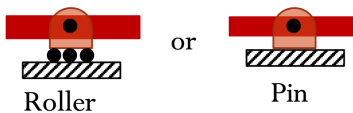
- Sign conventions:

$y'' < 0$, Concave down



Boundary Conditions

- ① Free Rotation $y = 0$
 $y' \neq 0$



- ② Fixed End $y = 0$
 $y' = 0$



- ③ Free End, None

- ④ Connection Point $y_1 = y_2$, $y'_1 = y'_2$

Calculation of $y(x)$

① Draw the free-body diagram

$$\sum \vec{F} = 0, \quad \sum \vec{M} = 0$$

Get reaction Forces, Moments

③ Get $M(x) \Rightarrow y'' = \frac{M(x)}{EI}$

④ Integration, $y' = \int y'' dx + a_0$

$$y = \int y' dx + a_1$$

② $V-x, M-x$ Diagram

对于外力 F, M (于某点)

$V_{\text{jump}} = F$ on the point

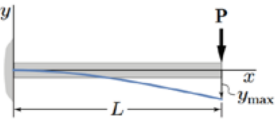
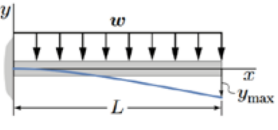
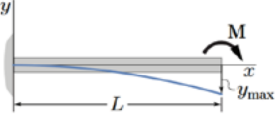
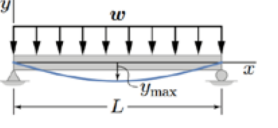
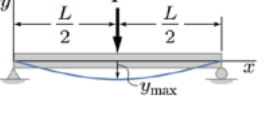
$M_{\text{jump}} = -M$ on the point

⑤ Apply Boundary Conditions

代入 BD

Solve $a_0, a_1 \dots$

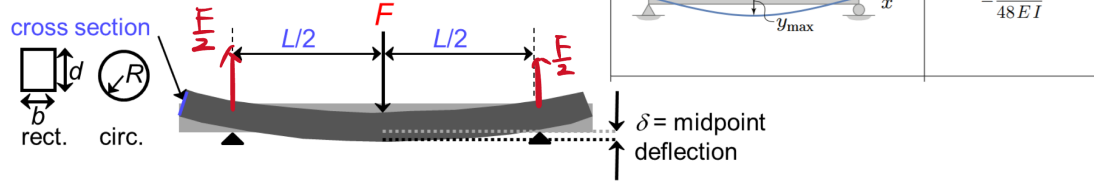
Method of Superposition 我不会 10\

Beam Deflection			
Diagram	Max. Deflection	Slope at End	Elastic Curve
	y_{max}	θ	$y(x)$
	$-\frac{PL^3}{3EI}$	$-\frac{PL^2}{2EI}$ $y'(end)$	$y(x) = \frac{P}{6EI} (x^3 - 3Lx^2)$
	$-\frac{wL^4}{8EI}$	$-\frac{wL^3}{6EI}$	$y(x) = -\frac{w}{24EI} (x^4 - 4Lx^3 + 6L^2x^2)$
	$-\frac{ML^2}{2EI}$	$-\frac{ML}{EI}$	$y(x) = -\frac{M}{2EI} x^2$
	$-\frac{5wL^4}{384EI}$	$\pm \frac{wL^3}{24EI}$	$y(x) = -\frac{w}{24EI} (x^4 - 2Lx^3 + L^3x)$
	$-\frac{PL^3}{48EI}$	$\pm \frac{PL^2}{16EI}$	For $0 \leq x \leq \frac{L}{2}$ $y(x) = \frac{P}{48EI} (4x^3 - 3L^2x)$



Application: Measure stiffness E for Brittle Materials

Instead, a 3 point bending test is often used



Stress at Failure:

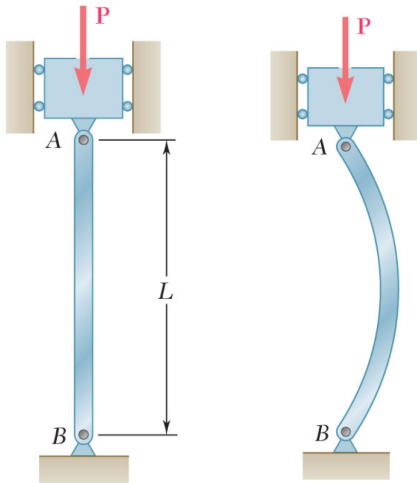
$$M = \frac{LF}{4}$$

$$\sigma_f = \frac{Mc}{I}, \text{ where } c = \Delta$$

Buckling of Columns

$\sigma < \sigma_{all} / \sigma_{ult}$ as buckles

INSTABILITY - CAUSED FAILURE



Euler's EQs for Pin-Ended Columns

$$EI y'' = M = -Py$$

$$y'' + \frac{P}{EI} y = 0$$

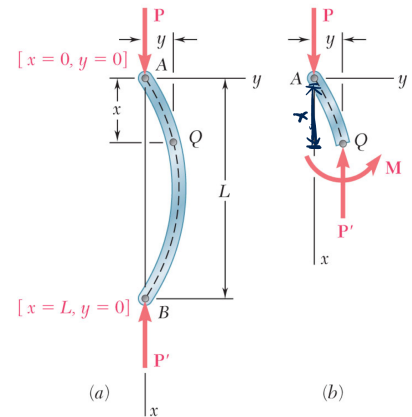
$$\frac{P}{EI} = \omega^2$$

$$y = A \cos(\omega x) + B \sin(\omega x)$$

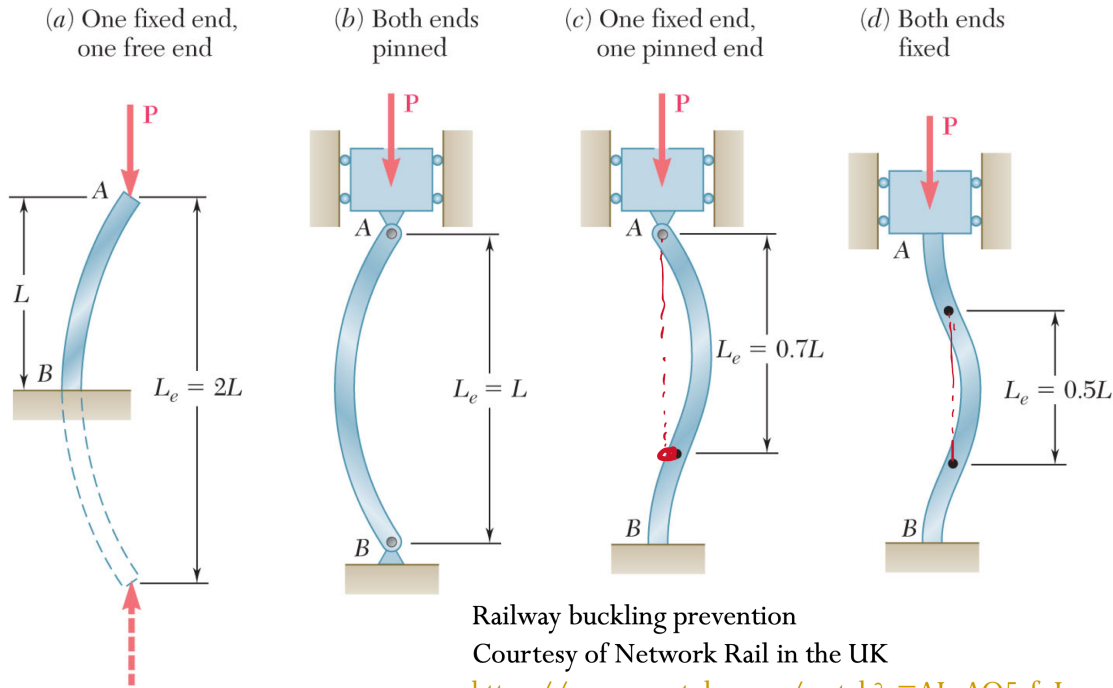
$$\text{IVP: } y(0) = 0, y(L) = 0$$

$$\Rightarrow P = \frac{n^2 \pi^2 EI}{L^2}, n \in \mathbb{Z}$$

$$P = \frac{\pi^2 EI}{L^2}$$



Extension of Euler's EQs

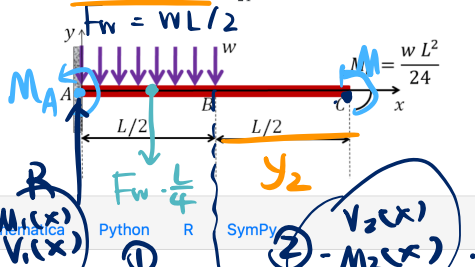


$y'_1(0) = 0$
 $y_1(0) = 0$

- ① Draw the free-body diagram
- $\sum \vec{F} = 0$, $\sum \vec{M} = 0$
 Get reaction Forces, Moments

HW11.8. Determine the slope and deflection at the free end

The cantilever beam AC with length $L = 7$ m has elasticity modulus $E = 216$ GPa, cross-section with moment of inertia $I = 43 \times 10^{-6} \text{ m}^4$ and supports a uniform distributed load of intensity $w = 6$ kN/m over half of its length and a couple moment $M_o = \frac{wL^2}{24}$ at the end C .



$w = 6$; % kN/m
 $L = 7$; % m
 $E = 216$; % GPa
 $Iz = 0.000043$; % m⁴

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Determine the deflection y_c at the tip of the beam (point C).

y_c number (rtol=0.01, atol=1e-05)

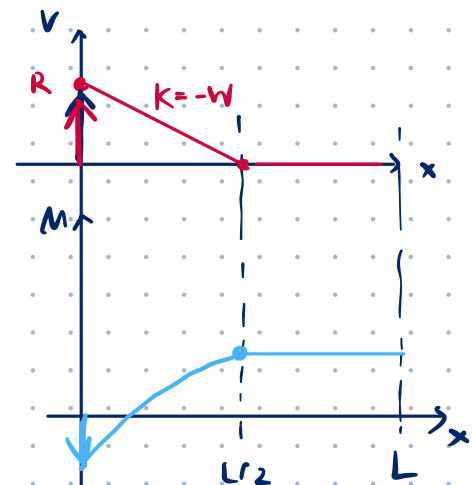
mm

Determine the slope θ_c at the tip of the beam (point C).

θ_c number (rtol=0.01, atol=1e-05)

rad

- ② $V-x$, $M-x$ Diagram
- 对于外力 F, M (于某点)
- $V_{\text{jump}} = F$ on the point
 $M_{\text{jump}} = -M$ on the point



```

from sympy import *
x, c1, c2, d1, d2 = symbols('x, c1, c2, d1, d2')
w = 9e3 # kN/m
l = 8 # m
E = 192e9 # GPa
I = 0.000044 # m^4
M = w*l**2/24
Fw = w*l/2
R = Fw

```

$$M = \frac{wl^2}{24}$$

```

V1 = -w*x + R
V2 = 0

```

```

# to get K to generate
M0 = Fw*l/4 - M

```

```

M1 = -M0 + integrate(V1, x)
M2 = M1.subs(x, l/2)

```

```

y1pp = M1/(E*I)
y2pp = M2/(E*I)

```

```

y1p = integrate(y1pp, x) + c1
y1 = integrate(y1p, x) + c2
y2p = integrate(y2pp, x) + d1
y2 = integrate(y2p, x) + d2

```

```

e1 = y1.subs(x, 0)
e2 = y1p.subs(x, 0)
e3 = y1.subs(x, l/2) - y2.subs(x, l/2)
e4 = y1p.subs(x, l/2) - y2p.subs(x, l/2)
ss = solve([e1, e2, e3, e4], [c1, c2, d1, d2])

```

```

y2s = y2.subs(d1, ss[d1]).subs(d2, ss[d2])
y2ps = y2p.subs(d1, ss[d1]).subs(d2, ss[d2])
y2s.subs(x, l)*1e3, atan(y2ps.subs(x, l))

```

③ Get $M(x) \Rightarrow y'' = \frac{M(x)}{EI}$

④ Integration, $y' = \int y'' dx + C_1$
 $y = \int y' dx + C_2$

⑤ Apply Boundary Conditions

$$\begin{aligned} y_1(0) &= 0 \\ y_1'(0) &= 0 \end{aligned} \quad \left\{ \begin{aligned} y_1\left(\frac{l}{2}\right) &= y_2\left(\frac{l}{2}\right) \\ y_1'\left(\frac{l}{2}\right) &= y_2'\left(\frac{l}{2}\right) \end{aligned} \right.$$

$$\Rightarrow C_1, C_2, d_1, d_2$$

⑥ Substitute

$$\Rightarrow \underline{y_1(x)}, \underline{y_2(x)}$$

$$\underline{y_c = y_2(l)}$$

$$\underline{\theta_c = \arctan(y_2'(l))}$$